

Hierarchical Multimodal Lecture-Video Topic Segmentation

A Low-Resource Benchmark and Diagnostic Study

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Problem

- Long lecture videos are hard to navigate without reliable chapters.
- Manual chapters are inconsistent, missing, or too coarse.
- Students need topic-level and subtopic-level access, not only video-level metadata.

- Large chaptering systems exist, but use thousands to hundreds of thousands of videos.
- Low-resource lecture segmentation lacks compact, auditable hierarchical benchmarks.
- LECSEG asks: what works when supervision is limited to 30 reviewed lecture videos?

Dataset Contribution

Videos	30
Duration	32.52 hours
Domains	Biology, CS, Math, Philosophy, Physics
Chapter boundaries	419
Reviewed subtopics	904

Pipeline

- ① Whisper transcripts and sentence timeline
- ② Sentence embeddings across multiple models
- ③ OCR, shot, and prosody extraction
- ④ Cross-model boundary evidence
- ⑤ Selector, hierarchy, and local LLM diagnostics

Main Result

Method	Pk	WD	F1@2
BGE-divisive baseline	0.3884	0.3956	0.0878
Cross-model conservative	0.3713	0.3764	0.0237
Balanced selector	0.3588	0.3739	0.0893

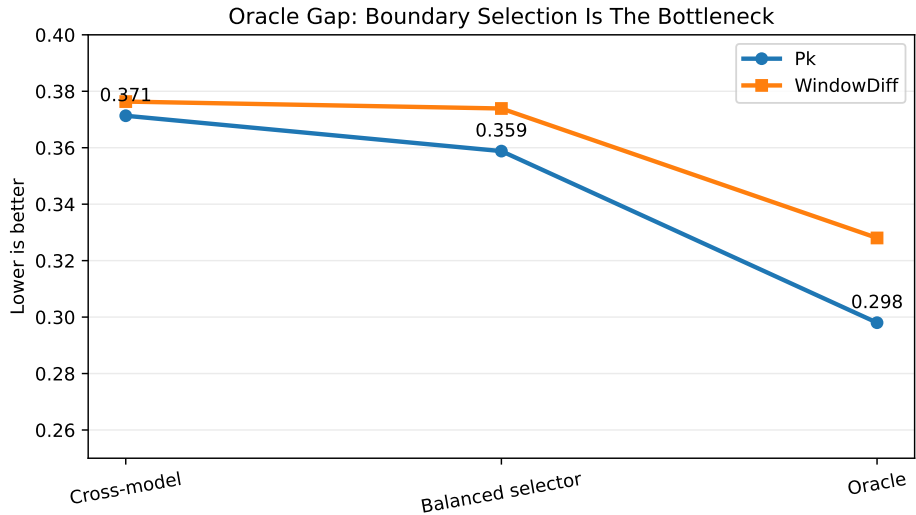
Best deployable operating point; not an external SOTA claim.

Same-Dataset TreeSeg Comparison

Method	Pk	WD	F1@2
Balanced selector	0.3588	0.3739	0.0893
TreeSeg-style MPNet	0.4320	0.4673	0.1733
TreeSeg-style E5-large	0.4322	0.4654	0.1576

TreeSeg-style splitting hits more exact boundaries but hurts segment consistency.

Oracle Gap



Case Studies

- Success: Physics videos where multimodal-grid variants reduce Pk/WD.
- Failure: selector over-switching damages Pk/WD on some videos.
- Domain weakness: Math needs better transcript/notation handling.

Evidence: docs/CASE_STUDIES.md

Compute Efficiency

- Classical and cross-model methods run locally after embeddings are cached.
- Selector sweep is lightweight compared with training large chaptering models.
- High-resource systems use orders of magnitude more videos.

Limitations

- 30-video seed benchmark, not a universal benchmark.
- Creator-provided chapters are useful but imperfect.
- Selector is not domain-general under leave-domain-out evaluation.
- Local LLM verifier is implemented but not promoted without full benchmark metrics.

- LECSEG contributes a reproducible low-resource hierarchical lecture benchmark.
- It achieves statistically supported local gains over implemented baselines.
- The main research insight is the oracle gap: robust boundary selection remains unsolved.